

ADM - TD1

Clément RENIERS

Table des matières

Ex1	2
Exercice 2	3

Ex1

1. $\forall i, m_i = 1$, donc

$$g(E) = \frac{\sum_{i=1}^n m_i a_i}{\sum_{i=1}^n m_i} = \frac{1}{8} \sum_{i=1}^8 a_i$$

D'où

$$\begin{aligned} g(E) &= \left(\frac{1}{8}\right)(3 + 0 + 1 + 0 + 4 + 2 + 4 + 2 \quad 0 + 1 + 1 + 2 + 2 + 3 + 3 + 4) \\ &= \left(\frac{1}{8}\right)(16 \quad 16) \\ &= (2 \quad 2) \end{aligned}$$

$$\begin{aligned} g(A_1) &= \left(\frac{1}{4}\right)(3 + 0 + 1 + 0 \quad 0 + 1 + 1 + 2) \\ &= \left(\frac{1}{4}\right)(4 \quad 4) \\ &= (1 \quad 1) \end{aligned}$$

$$\begin{aligned} g(A_2) &= \left(\frac{1}{4}\right)(4 + 2 + 4 + 2 \quad 2 + 3 + 3 + 4) \\ &= \left(\frac{1}{4}\right)(12 \quad 12) \\ &= (3 \quad 3) \end{aligned}$$

Ainsi $G = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$

2. $\hat{X}_{i,j} = X_{i,j} - g_j$, donc

$$\hat{X} = \begin{pmatrix} 1 & -2 \\ -2 & -1 \\ -1 & -1 \\ -2 & 0 \\ 2 & 0 \\ 0 & 1 \\ 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\hat{Y} = \begin{pmatrix} 2 & -1 \\ -1 & 0 \\ 0 & 0 \\ -1 & 1 \end{pmatrix}$$

$$\hat{Z} = \begin{pmatrix} 1 & -1 \\ -1 & 0 \\ 1 & 0 \\ -1 & 1 \end{pmatrix}$$

3. $T = \hat{X}^T D \hat{X} = \hat{X}^T \hat{X} =$

$$W = M(A_1) + M(A_2) = \begin{pmatrix} 6 & -3 \\ -3 & 2 \end{pmatrix} + \begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix} = \begin{pmatrix} 10 & -5 \\ -5 & 4 \end{pmatrix}$$

$$B = G^T D_g G = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$$

4.

5. (j'ai oublié de prendre le TD hihi)

6.

$$\frac{1}{2} u^T B u = \frac{1}{2} (1 \ 1) \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 16$$

d'où

$$\begin{aligned} p &= \frac{u^T B u}{u^T W u} \\ &= \frac{16}{18} \\ &\approx 0.89 \end{aligned}$$

7.

$$\begin{aligned} I_b(E) &= \sum_{i=1}^n m_i \cdot d^2(a_i, b) \\ &= d^2((3, 0), c) + d^2((0, 1), c) + d^2((1, 1), c) + d^2((0, 2), c) + d^2((4, 2), c) + d^2((2, 3), c) + d^2((4, 3), c) + d^2((2, 4), c) \\ &= 110 \end{aligned}$$

Exercise 2

1.

$$\begin{aligned} V_1 &= \frac{1}{6} (1 \ 0 \ 1) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \\ &\vdots \\ V &= \frac{1}{6} \begin{pmatrix} 4 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{pmatrix} \end{aligned}$$

2.

$$\begin{aligned} I_O(E) &= \sum_{i=1}^n m_i \cdot d^2(a_i, O) \\ &= \frac{1}{6} (d^2((1, 0, -1), O) + d^2((0, 1, -1), O) + d^2((-1, 1, 0), O) \\ &\quad + d^2((0, -1, 1), O) + d^2((-1, 0, 1), O) + d^2((1, -1, 0), O)) \\ &= \frac{1}{6} (2 + 2 + 2 + 2 + 2 + 2) \\ &= 2 \end{aligned}$$

3.

$$\begin{aligned} \chi_A &= \det(V - \lambda I_3) \\ &= \frac{1}{3} \begin{vmatrix} 2 - \lambda & -1 & -1 \\ -1 & 2 - \lambda & -1 \\ -1 & -1 & 2 - \lambda \end{vmatrix} \\ &= -\frac{1}{3} \lambda (-3 + \lambda) \end{aligned}$$

BALC ça fait $\{0, 1, 1\}$

4. $u = k \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

5. $u' = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

6. $\begin{pmatrix} \sqrt{2} & 0 \\ -\frac{\sqrt{2}}{2} & -\sqrt{\frac{3}{2}} \\ -\frac{\sqrt{2}}{2} & -\sqrt{\frac{3}{2}} \\ -\frac{\sqrt{2}}{2} & \sqrt{\frac{3}{2}} \\ -\sqrt{2} & 0 \\ \frac{\sqrt{2}}{2} & \sqrt{\frac{3}{2}} \end{pmatrix}$